

Voltra EV-60 Electric Delivery Van

Operator & Fleet Maintenance Manual

Sample document: this is a fictional vehicle created to demonstrate multi-model retrieval in the Fleet Assistance Copilot. Values are realistic but not taken from any real OEM.

Document Control

Field	Value
Document ID	VM-EV60-OM
Vehicle model	EV-60
Variants covered	EV-60 SR (45 kWh), EV-60 LR (60 kWh)
Vehicle category	N1 light goods vehicle (battery electric)
Region	India
Version	2.0
Effective date	2026-08-15

1. About This Manual

This manual covers operation, charging, battery care, preventive maintenance and telematics thresholds for the Voltra EV-60 battery-electric delivery van. The EV-60 has no engine oil, no fuel system and no exhaust after-treatment, so several items in the generic Fleet Operations Handbook (engine oil change, DPf, AdBlue, fuel theft) do not apply to it.

The EV-60 is used for urban and peri-urban last-mile deliveries with daily routes of 80–200 km.

2. Vehicle Specifications

Table 2.1 – EV-60 technical specifications

Specification	EV-60 SR	EV-60 LR
Battery	45 kWh LFP	60 kWh LFP
Rated range (MIDC)	210 km	280 km
Realistic laden city range	150–170 km	200–230 km
Motor	PMSM, 90 kW peak / 60 kW continuous	PMSM, 90 kW peak / 60 kW continuous
Peak torque	300 Nm	300 Nm
Gross vehicle weight (GVW)	3,500 kg	3,500 kg
Payload	1,300 kg	1,250 kg
Cargo volume	10.5 m³	10.5 m³
Top speed	90 km/h (80 km/h in fleet mode)	90 km/h (80 km/h in fleet mode)
AC charging (on-board charger)	11 kW, 0–100% in about 5 h	11 kW, 0–100% in about 6.5 h
DC fast charging (CCS2)	50 kW, 20–80% in about 35 min	60 kW, 20–80% in about 40 min
Tyre size	215/75 R16C	215/75 R16C

3. Fluid and Consumable Capacities

Table 3.1 – EV-60 fluid capacities

System	Capacity	Specification
Battery and motor coolant loop	9 L	Low-conductivity EV coolant (OEM spec VTR-C2)
Reduction gear oil	1.8 L	SAE 75W-85 synthetic
Brake fluid	0.9 L	DOT 4
Engine oil	Not applicable	The EV-60 has no engine
12 V auxiliary battery	70 Ah	AGM type

4. Tyre Specifications and Pressures

Table 4.1 – EV-60 tyre pressures

Position	Cold pressure (unladen)	Cold pressure (laden)
Front axle	4.5 bar (65 psi)	4.5 bar (65 psi)
Rear axle	4.5 bar (65 psi)	5.0 bar (73 psi)

Because the battery makes the EV-60 heavier than a diesel van of the same size, under-inflation reduces range noticeably: 0.5 bar under-inflation reduces range by about 3–4%.

5. Battery Care and Charging

The EV-60 uses lithium iron phosphate (LFP) cells, which tolerate full charging. For daily use, charge to 90%. Charge to 100% at least once a week so the battery management system can recalibrate the state-of-charge (SoC) estimate.

Table 5.1 – EV-60 state-of-charge bands

State of charge (SoC)	Status	What happens / driver action
90–100%	Full	Normal. Regenerative braking is reduced above 95%.
20–90%	Normal operating band	Normal driving.
15–20%	Plan charging	Plan a charging stop on the current route.
7–15%	Low SoC warning (amber)	Telematics alert raised; drive directly to a charger.
Below 7%	Critical (turtle mode)	Power limited to about 30 kW and top speed to 40 km/h; charge immediately.

Table 5.2 – EV-60 battery temperature limits

Battery temperature	Status	Effect
15–40 °C	Normal	Full power and full charging speed.
0–15 °C	Cold	DC charging speed reduced; precondition while plugged in.
40–50 °C	Hot	Charging speed reduced to protect cells.
≥ 50 °C	Warning	Amber battery-temperature lamp; power derate begins.
≥ 55 °C	Critical	Red lamp; stop safely and switch off. Contact the EV support desk.

5.1 Charging Procedure

1. Park, select P and switch the vehicle off.
2. Inspect the charging cable and connector for damage or water; never use a damaged cable.
3. Open the charge-port flap and plug in firmly until the connector latches.
4. Confirm the charging lamp shows steady green on the charge port.
5. To stop, end the session on the charger or app, then unlatch and remove the connector.
6. Close the charge-port flap before driving.

Limit DC fast charging to 2 sessions per day where possible. Frequent back-to-back DC fast charging raises battery temperature and accelerates degradation. Battery warranty: 8 years or 160,000 km, or until state of health (SOH) falls below 70%.

6. Regenerative Braking

Table 6.1 – EV-60 regenerative braking levels

Regen level	Deceleration when lifting off	Recommended use
L0 – Off	Coasting	Long highway descents only when battery is above 95%
L1 – Low	Up to 0.08 g	Highway cruising
L2 – Medium (default)	Up to 0.14 g	Mixed driving
L3 – High (one-pedal)	Up to 0.20 g	Stop-and-go city delivery routes

Regenerative deceleration up to 0.20 g is not counted as a harsh braking event by the telematics platform.

7. Instrument Cluster Warning Lights

Table 7.1 – EV-60 warning lamps

Warning lamp	Colour	Meaning	Driver action
EV system fault	Red	High-voltage system fault detected	Stop safely, switch off, do not restart. Call EV support.
Battery over-temperature	Red	Battery at or above 55 °C	Stop safely and switch off; keep away from the vehicle if smoke or hissing is present.
Low state of charge	Amber	SoC below 15%	Drive directly to a charger.
Reduced power (turtle)	Amber	SoC below 7% or thermal derate	Charge immediately; top speed limited to 40 km/h.
12 V charging fault	Amber	DC-DC converter not charging the 12 V battery	Book inspection; the vehicle may not restart.
AVAS fault	Amber	Pedestrian warning sound not working	Drive with extra care at low speed; book inspection.
READY	Green	High-voltage system active; vehicle can move silently	Information only. Always select P before leaving the seat.

8. Common Diagnostic Trouble Codes (DTC)

Table 8.1 – Frequent EV-60 DTCs

DTC	Description	Severity	Recommended action
P0AA6	High-voltage isolation fault	Critical	Stop; do not touch orange HV cables; tow to an authorised EV workshop.
P0A0A	High-voltage interlock circuit open	Critical	Vehicle will not enter READY; tow to EV workshop.

DTC	Description	Severity	Recommended action
P0A7F	Battery pack deterioration	Medium	Schedule a state-of-health test.
P0C73	Motor electronics coolant pump fault	High	Stop when safe; power derate likely; inspect coolant level.
U0293	Lost communication with EV powertrain control module	High	Restart once; if it persists, tow to workshop.
P0D2B	On-board charger input voltage out of range	Medium	Try another charger; report the charging location.

9. Preventive Maintenance Schedule

Service at the kilometre or time interval, whichever comes first. There is no engine oil change on the EV-60.

Table 9.1 – EV-60 preventive maintenance schedule

Interval	Service items
Daily (pre-trip)	SoC check, tyre visual check, charging cable and port inspection, lights, AVAS sound check
15,000 km or 12 months	Brake inspection, cabin air filter, tyre rotation, 12 V battery test, HV isolation check via diagnostics
30,000 km or 24 months	Brake fluid replacement, coolant level and conductivity check, suspension inspection
60,000 km or 48 months	Reduction gear oil replacement, coolant replacement
Every 12 months	Battery state-of-health (SOH) report from the battery management system

10. Telematics Thresholds for EV-60

Electric motors deliver full torque instantly, so harsh acceleration events are more common on the EV-60 than on diesel vans. Thresholds are set accordingly.

Table 10.1 – EV-60 telematics thresholds

Signal / event	Warning threshold	Critical threshold
Harsh braking (friction brakes)	Deceleration ≥ 0.40 g	Deceleration ≥ 0.55 g
Harsh acceleration	Acceleration ≥ 0.35 g	Acceleration ≥ 0.45 g
Harsh cornering	Lateral ≥ 0.35 g	Lateral ≥ 0.45 g
Overspeed	> 80 km/h for 10 s	> 90 km/h
Energy consumption	> 260 Wh/km (trip average)	> 320 Wh/km
State of charge	$< 15\%$	$< 7\%$
Battery temperature	≥ 50 °C	≥ 55 °C
DC fast-charge sessions	> 2 per day	> 4 per day

11. Energy Efficiency Guidance

Table 11.1 – EV-60 expected energy consumption

Condition	Expected consumption	Notes
City, unladen	170–190 Wh/km	Regen L3
City, full payload	200–230 Wh/km	Regen L3
Highway 70–80 km/h, full payload	230–260 Wh/km	Aerodynamic drag dominates

Condition	Expected consumption	Notes
AC on (any condition)	+8–12%	Set cabin to 24 °C

- Precondition the cabin while plugged in instead of on battery.
- Use regen level L3 on stop-and-go routes.
- Keep speed at or below 70 km/h on highways; consumption rises sharply above 80 km/h.
- Keep tyres at the laden pressure when carrying more than half payload.

12. High-Voltage Safety and Emergency Procedures

12.1 High-Voltage Safety

The traction battery operates at about 350 V DC. All high-voltage cables are orange. Drivers must never open covers marked with the high-voltage symbol. Only EV-certified technicians may operate the manual service disconnect.

12.2 Accident Involving an EV-60

1. Select P, switch the vehicle off and remove the key fob to at least 5 m away.
2. Switch on hazard lights and evacuate occupants.
3. Tell emergency services that the vehicle is a battery-electric van.
4. Do not touch damaged orange cables or leaking fluids.
5. If there is smoke, hissing or popping from the battery, move everyone at least 15 m away and upwind.

12.3 Battery Fire

A lithium battery fire can re-ignite hours after it appears to be out. Do not attempt to fight a battery fire with a hand-held extinguisher; evacuate and call the fire service. After any battery fire or deep water immersion, the vehicle must be quarantined in the open, at least 15 m from other vehicles and buildings, for 48 hours.

12.4 Water Wading

Maximum wading depth is 300 mm at walking pace. Do not drive through water deeper than 300 mm.

13. Frequently Asked Questions

Q: Does the EV-60 need an engine oil change? A: No. The EV-60 has no engine; only reduction gear oil is replaced, every 60,000 km or 48 months.

Q: How far should I charge the EV-60 every day? A: Charge to 90% daily and to 100% once a week for SoC calibration.

Q: Why did my EV-60 slow down to 40 km/h? A: Turtle mode activates below 7% SoC or when the battery is too hot; charge the vehicle or let it cool.